It is recommended that you spend about 30 minutes on this question.

2 Modern day lifts in tall buildings are of a relatively simple design. A passenger car is connected to a counterweight by a steel cable which passes over a system of pulleys at the top of the lift shaft, as shown in Fig. 2.1.

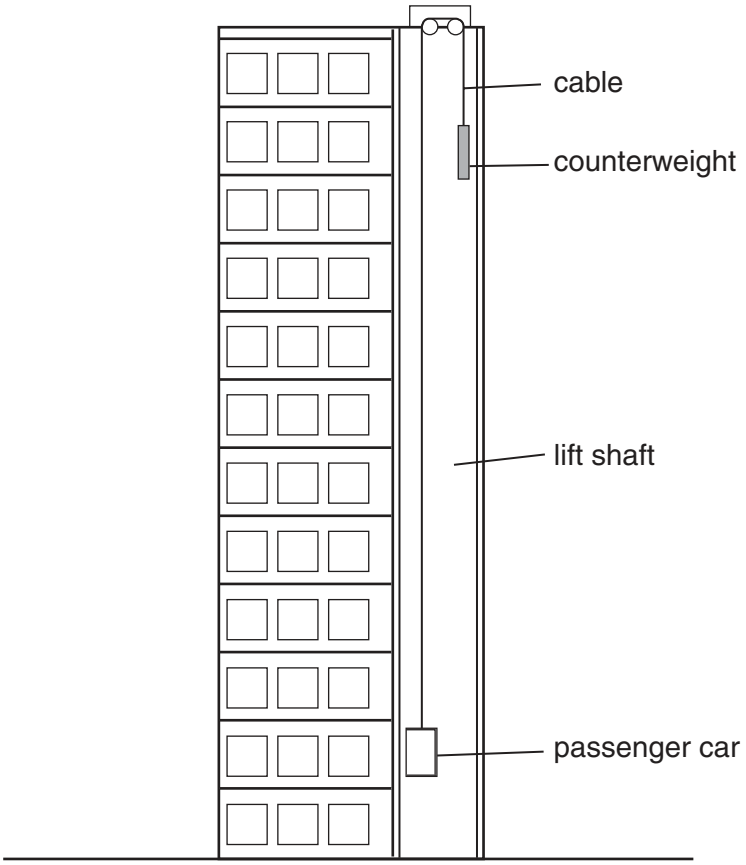

Fig. 2.1

The passenger car and the counterweight are of similar mass. A motor, which is connected to the pulley system, is used to raise and lower the passenger car. A braking system enables the passenger car to come to rest at any desired position. If the motor and the braking system were to fail, the passenger car and counterweight would move under the action of gravity until one of several safety features incorporated in the design of the system (not shown on Fig. 2.1) brings the passenger car and counterweight to rest.

A failure of the motor and braking system may be simulated in the laboratory by connecting two objects A and B of mass m_A and m_B respectively by a thin steel wire which passes over a pulley, as shown in Fig. 2.2.

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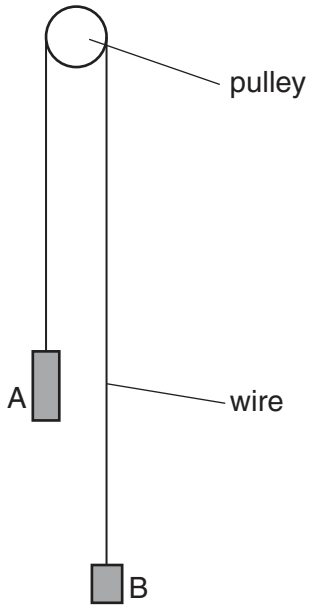

Fig. 2.2

Design a laboratory experiment to investigate how the acceleration of object A depends on the mass m_B of object B when the system is allowed to move freely under the action of gravity. You may assume that m_A is constant throughout and that $m_A > m_B$. You should draw a diagram showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements that would be taken,

(c) how the acceleration of object A would be determined,

(d) any safety precautions that you would take.

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